MODULE 08 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Hybrid Connectivity and Data Synchronisation

Links that survive failure, data that stays in step, and legacy systems that keep talking.

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Hybrid connectivity is designed for the day a link fails, not the day it works.

Direct Connect or Site-to-Site VPN?

DIRECT CONNECT

Private, predictable

- 1, 10, 100 or 400 Gbps dedicated ports
- Consistent latency, no internet path
- Weeks to provision
- Not encrypted by default

SITE-TO-SITE VPN

Fast, encrypted

- 1.25 Gbps per tunnel; 5 Gbps with large tunnels
- Latency follows the internet
- Up in minutes
- IPsec encrypted by default

THE TIEBREAKER

Needed this week means VPN. Sustained throughput and predictability mean Direct Connect. Most answers use both.

The three virtual interface types

VIF	CONNECTS TO	USE IT FOR
Private	A virtual private gateway or DX gateway	Reaching VPCs on private addresses
Public	AWS public endpoints	S3, DynamoDB and public services over the circuit
Transit	A Direct Connect gateway only	Reaching a Transit Gateway and everything behind it

ON THE EXAM

Direct Connect and Transit Gateway in the same question means a transit VIF through a DX gateway.

Direct Connect resiliency models

MODEL	SHAPE	FOR
Maximum	Separate connections on separate devices at two or more locations	Critical production
High	One connection at each of two locations	Most production
Development and Test	Two connections, separate devices, one location	Non-critical work
Classic	Created without the toolkit; no resiliency	Existing single links

ON THE EXAM

Surviving the loss of a whole location needs at least High. Surviving that plus a device failure needs Maximum.

VPN numbers that decide answers

- **Two tunnels per connection,** in different AWS endpoints, for redundancy.
- **A virtual private gateway caps near 1.25 Gbps** in aggregate, however many connections.
- **Large bandwidth tunnels: 5 Gbps,** since November 2025. Transit Gateway or Cloud WAN only.
- **ECMP** spreads traffic across tunnels for more bandwidth. Needs BGP and a Transit Gateway.
- **Direct Connect is preferred over VPN** for the same prefix, so failover is automatic.

> ON THE EXAM
>
> Any VPN throughput above about 1.25 Gbps needs a Transit Gateway. The virtual private gateway is the bottleneck.

Encrypting traffic over Direct Connect

MACSEC

Encrypt the circuit

- Supported dedicated ports only
- Line rate, no throughput penalty
- Your router to the AWS router
- Not on hosted connections

PRIVATE IP VPN

Encrypt the journey

- IPsec over a transit VIF
- Limited by tunnel bandwidth
- End to end, whatever is in between
- Needs a Transit Gateway

THE TIEBREAKER

Line-rate encryption on your own dedicated port means MACsec. End-to-end or hosted means private IP VPN.

A resilient hybrid path

01
On premises
Two routers, two Direct Connect locations.

02
Transit VIFs
One per connection, into a DX gateway.

03
Transit Gateway
Hub for VPCs in many accounts.

04
VPN backup
Large tunnels to the same Transit Gateway.

05
BGP decides
DX preferred; VPN takes over on failure.

ON THE EXAM

Two DX locations for capacity, a VPN for the worst case, one Transit Gateway so routing stays simple.

Keeping data in step across the boundary is a different problem from moving it once.

Hybrid data services

SERVICE	WHAT IT DOES
AWS DataSync	Scheduled, incremental copy with validation; agentless from other clouds
S3 File Gateway	NFS or SMB share on premises, files stored as S3 objects, local cache
Volume Gateway	iSCSI block volumes, backed up as EBS snapshots
Tape Gateway	Virtual tape library for existing backup software
AWS Transfer Family	Managed SFTP, FTPS, FTP and AS2 endpoints on S3 or EFS
FSx File Gateway	Closed to new customers since October 2024

ON THE EXAM

DataSync moves data on a schedule. Gateways give on-premises applications a local protocol that lands in AWS.

DataSync or S3 File Gateway?

CHOOSE DATASYNC WHEN

Copy on a schedule

- Existing file system stays as it is
- Data needed in AWS hourly or daily
- Integrity validation and throttling
- Also AWS to AWS and cloud to AWS

CHOOSE S3 FILE GATEWAY WHEN

Applications write to S3

- Applications keep an NFS or SMB share
- Each file becomes an S3 object
- Local cache for fast recent reads
- Continuous, not batch

THE TIEBREAKER

Applications must keep writing to a share, and files must land in S3 continuously: File Gateway.

Integrating legacy systems

- **Amazon MQ** runs ActiveMQ or RabbitMQ: JMS, AMQP, STOMP and MQTT apps move unchanged.
- **API Gateway REST mapping templates** reshape requests and responses, such as XML to JSON.
- **Lambda as an adapter** between a legacy interface and a modern one.
- **EventBridge Pipes** filter, enrich and transform between a source and a target.
- **Transfer Family AS2** for business-to-business file exchange.

> ON THE EXAM
>
> Skill 1.3.3 says data transformations. The legacy side does not change; the integration layer translates.

Amazon MQ or SQS and SNS?

CHOOSE AMAZON MQ WHEN

Existing broker protocols

- JMS, AMQP, STOMP, MQTT or OpenWire
- Messaging code cannot change
- Lift and shift of ActiveMQ or RabbitMQ
- Broker instances you size

CHOOSE SQS AND SNS WHEN

Cloud-native messaging

- New or rewritten applications
- AWS SDK and HTTPS APIs
- Virtually unlimited scale
- Nothing to size or patch

THE TIEBREAKER

Cannot change the messaging code means Amazon MQ. New code or a rewrite means SQS and SNS.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Connectivity needed this week	Site-to-Site VPN now, Direct Connect later
Survive losing a location and a device	Maximum resiliency
VPN backup must carry several Gbps	Transit Gateway with large tunnels
Line-rate encryption over 100 Gbps DX	MACsec
Apps keep writing to a share; files in S3	S3 File Gateway
JMS application, code cannot change	Amazon MQ

ON THE EXAM

Every connectivity question has a failure in it. Find it before reading the options.

> Meridian Freight is opening a new distribution centre that must connect to its VPCs, through an existing Transit Gateway, within a week. In steady state it will send about 3 Gbps of predictable traffic.
>
> Ordering a Direct Connect connection at the nearest location takes about six weeks.

What should the solutions architect recommend?

A. Order Direct Connect and connect the site when it is ready

B. A Site-to-Site VPN to the Transit Gateway now, then Direct Connect, keeping the VPN as backup

C. A Site-to-Site VPN to a virtual private gateway in each VPC

D. AWS Client VPN for the distribution centre's users

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A. Wait for Direct Connect

Misses the one-week deadline by five weeks.

B. VPN now, DX later, VPN as backup

Meets the deadline, then the steady state, and keeps a failover path.

C. VPN per VPC

Ignores the existing Transit Gateway, and each VGW caps near 1.25 Gbps.

D. Client VPN

Connects individual users, not a site's network.

THE SIGNAL

A deadline shorter than the DX lead time means VPN first. Predictable sustained traffic means DX after.

> Meridian Freight's order platform depends on Direct Connect and is classed as business-critical. The company must keep full Direct Connect bandwidth even if an entire Direct Connect location fails, and a device fails at the same time elsewhere.
>
> A VPN backup is not acceptable for this traffic.

Which Direct Connect design meets these requirements?

A Two connections on separate devices at one location

B One connection at each of two locations

C Separate connections on separate devices at two or more locations

D One Direct Connect connection with a Site-to-Site VPN backup

◆ PAUSE AND DECIDE

Which Direct Connect design meets these requirements?

A. Two devices, one location

Development and Test model. Losing the location loses everything.

B. One per location

High model. Survives a location, but not a device failure at the survivor.

C. Separate devices, two-plus locations

Maximum model. Survives the location and a device, at full bandwidth.

D. DX with VPN backup

The scenario rules out VPN as the backup.

THE SIGNAL

Location failure plus device failure, with no drop in bandwidth, is the definition of Maximum resiliency.

> Meridian Freight's Direct Connect carries about 4 Gbps. Its backup is a Site-to-Site VPN on a virtual private gateway. During a recent circuit failure the VPN carried only a fraction of the traffic.
>
> The backup must carry the full 4 Gbps in a future failure.

What should the solutions architect do?

A. Move the VPN to a Transit Gateway and use large bandwidth tunnels

B. Add three more VPN connections to the virtual private gateway

C. Enable acceleration on the existing VPN connection

D. Replace the customer gateway with a faster appliance

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Transit Gateway, large tunnels

Removes the VGW ceiling; one 5 Gbps tunnel covers 4 Gbps.

B More VPN connections on the VGW

The VGW's aggregate cap near 1.25 Gbps applies however many you add.

C Accelerated VPN

Improves the internet path, not the gateway's ceiling. Also not supported with large tunnels.

D Faster customer gateway

The bottleneck is on the AWS side.

THE SIGNAL

The virtual private gateway is the ceiling. More tunnels into it change nothing.

> Meridian Freight replicates its trading data over a dedicated 100 Gbps Direct Connect connection. A new regulation requires all traffic over the circuit to be encrypted.
>
> The link runs near capacity at month end, and the encryption must not reduce throughput.

Which solution meets these requirements?

A. A private IP VPN over a transit VIF

B. TLS in every application that uses the link

C. A Site-to-Site VPN over a public VIF

D. MACsec on the dedicated connection

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A Private IP VPN

Encrypts, but tunnel bandwidth falls far short of 100 Gbps.

B TLS everywhere

Depends on every application, so it cannot guarantee all traffic is covered.

C VPN over a public VIF

Same tunnel bandwidth limit, with a more complex path.

D MACsec

Encrypts the circuit at line rate on a supported dedicated port.

THE SIGNAL

All traffic, at 100 Gbps, with no throughput loss, on a dedicated port. Only MACsec fits all four.

> Meridian Freight's depot scanners save delivery images to an on-premises NFS share. The images must appear in Amazon S3 for analytics within minutes, and depot applications frequently reopen recent images.
>
> The depot applications cannot be changed.

Which solution should the solutions architect recommend?

A. An AWS DataSync task that runs every hour

B. AWS Transfer Family with an SFTP endpoint

C. Volume Gateway in cached mode

D. S3 File Gateway presenting an NFS share to the applications

◆ PAUSE AND DECIDE

Which solution should the solutions architect recommend?

A Hourly DataSync

Up to an hour late, which misses within minutes.

B Transfer Family SFTP

The applications would have to change to upload by SFTP.

C Volume Gateway

Presents block storage over iSCSI, not an NFS share, and stores snapshots, not objects.

D S3 File Gateway

Same NFS protocol, files land as S3 objects, cache serves recent reads.

THE SIGNAL

Unchanged NFS applications, files in S3 within minutes, fast local rereads. That is File Gateway's description.

> Meridian Freight is moving a legacy order-processing system to AWS. Its components exchange messages through an Apache ActiveMQ broker using JMS, with queues and topics.
>
> The messaging code cannot be modified, and the company wants the LEAST operational overhead.

What should the solutions architect use for messaging?

A. Amazon SQS through the SQS Java Messaging Library

B. A self-managed ActiveMQ cluster on EC2

C. Amazon MQ for ActiveMQ

D. Amazon EventBridge with a custom event bus

◆ PAUSE AND DECIDE

What should the solutions architect use for messaging?

A SQS with the JMS library

Requires changing client configuration and code, and offers queues only, not topics.

B Self-managed ActiveMQ

Compatible, but the company patches, scales and fails over the broker itself.

C Amazon MQ for ActiveMQ

The same broker, managed. Clients connect over JMS exactly as before.

D EventBridge

Uses AWS APIs, not JMS. The code would have to be rewritten.

THE SIGNAL

Code cannot change rules out every AWS-API service. Least overhead rules out running the broker yourself.

What to carry out of this module

- **VPN for speed, Direct Connect for steady state,** VPN kept as backup.
- **Count the failures:** location means High; location plus device means Maximum.
- **VPN above 1.25 Gbps needs a Transit Gateway.** Large tunnels reach 5 Gbps.
- **MACsec** for line-rate circuit encryption; **private IP VPN** for end to end.
- **File Gateway** for unchanged file apps; **Amazon MQ** for unchanged JMS or AMQP apps.

> ON THE EXAM
>
> Stale facts: VPN tunnels are no longer capped at 1.25 Gbps, and FSx File Gateway is closed to new customers.